Yashowardhan Rai

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Student member of the LIGO India Science Collaboration since 2025

Education

- **B.Tech., Engineering Physics**, Indian Institute of Technology, Bombay *('22 - '26)*
  GPA : 9.01/10
- **MSc. Physics with Specialisation in Astrophysics**, Indian Institute of Technology, Bombay *('26 - '27)*
  GPA : 9.01/10
  Thesis Title : "Unlocking and Applying Template Matching Polarimetry inside 3ML framework for Astrosat-CZTI"

Research Interests

Time Domain Astronomy - Follow Up and Modeling of Transients, Multi-Messenger Science, Compact Objects
Astronomical Software - Bayesian Methods, Simulations, Instrumentation-Modeling, Open Source Development

Publications and Conferences

- S. Mate, H. Li, U. Pathak, **Y. Rai** et. al. , "PolPy: A universal tool for X-ray/gamma-ray polarimetry using common data format standards", submitted to *Astrophysical Journal*
- U. Pathak, S. Patil, H. Tanenia, **Y. Rai** et. al." Can We Find the Emission Mechanism Behind the Extremely Bright GRB 230812B?", submitted to Astrophysical Journal
- R. Kumar, **Y. Rai** et al., "Lunar geochemistry from X-ray line flux ratios using CLASS on Chandrayaan 2", Accepted in Planetary Science Journal

Manuscript In Preparation :

- **Y. Rai** et. al., "Detection of Inverse Compton Emission From GRB260226A Using AstroSat-CZTI Polarimetry"
- **Y. Rai** et. al., "Non-Equilibrium Radiative Transfer Simulations to Probe the Ammonia Hyperfine Intensity Anomaly"
- U. Pathak, **Y. Rai** et. al. "Polarimetric Analysis of GRBs in 11 year Polarization Catalog of Astrosat-CZTI"
- U. Pathak, **Y. Rai**, "Joint Polarization Analysis of GRBs, Co-Detected by CZTI and POLAR"

Talks and Posters

- "Can Selective Photon Trapping Explain Ammonia Hyperfine Anomalies?", invited online talk at NRAO Socorro.
- Won best poster for "PolPy: A universal tool for hard x-ray polarimetry" at IITB's symposium of physics.
- R. Kumar, **Y. Rai** et al., "Lunar geochemistry from X-ray line flux ratios using CLASS on Chandrayaan 2", poster accepted at Lunar and Planetary Science Conference hosted at Lunar and Planetary Institute.

Research Projects

Hard X-ray Polarimetry Software: PolPy, CZTI and Daksha *(Dec 2023 - Jan 2025)*

Guide : Prof. Varun Bhalerao | Department of Physics, IITB | PolPy Paper

- Co-developed PolPy as a plugin for **Multi-Mission Maximum Likelihood Framework** and standardised Compton polarimetry routines enabling **joint-polarization measurements** using current and future polarimeters.
- Built an **injection-recovery pipeline** for the next-generation GRB polarimeter **Daksha**, coordinating file formats and coordinate conventions with the **POLAR** mission to synthesise GRB data for the testing of PolPy.
- Converted the **AstroSat-CZTI polarization pipeline** from legacy IDL to Python and added **pixel charge-sharing effects** to the response, thus enabling CZTI's **11-year catalogue of 40+ GRBs** analysis.
- Implemented CI/CD routines for PolPy, accelerated response-interpolation routines and **profiled, parallelised and optimised** the Daksha and CZTI pipelines thus making analysis at catalogue scale practical.

Bayesian Spectro-Polarimetry of Gamma-Ray Bursts *(Aug 2024 - present)*

Guide : Prof. Varun Bhalerao | Department of Physics, IITB | MSc Thesis

- **Led the spectro-polarimetric analysis of GRB260226A** with AstroSat-CZTI, building joint GBM-LAT-CZTI spectro-polarimetric fits in 3ML and compared to predictions of **Two-Zone Inverse Compton Emission.**
- Wrote **time-resolved spectral fitting scripts** and prompt emission light curves for GRB230812B, estimating the **photospheric radius** from the thermal (blackbody) component.
- Modelled the **multi-band afterglow** of GRB 230812B with **MCMC sampling of afterglowpy models**, constraining jet and circumburst parameters from the full set of light curves.
- Reported joint-polarization measurements of GRBs 161218B and 170202B, Co-Detected by CZTI and POLAR.

Non-LTE Modelling of the Ammonia Hyperfine Intensity Anomaly *(May 2025 - present)*

Guides : Dr. Thushara G.S. Pillai and Dr. Brian Svoboda | MIT Haystack Observatory and NRAO Socorro

- Used the **3-D radiative transfer code** Magritte to create synthetic images of NH_3 **hyperfine inversion transitions** in **cold protostellar cores**, validating against **exact LTE limits** and **known HCN anomalies**.
- Utilised the **latest NH_3-H_2 collisional rates**, from the LAMDA database, to demonstrate using Non-LTE simulations that **selective photon trapping** produces ammonia's **hyperfine intensity anomaly**.
- Built a χ^2-based retrieval to infer **cloud density, NH_3 abundance and Jeans radius** purely from line ratios and verified the formalism against Effelsberg 100m Telescope data of anomalous ammonia sources.
- **Scaled the retrieval to ∼2000 real sources** from the **Green Bank Ammonia Survey** (Orion A) and **Parkes telescope** data, enabling population-level study of anomalous cores.

Likelihood-Free Parameter Estimation of Gravitational Waves *(May 2025 - present)*

Guide : Prof. Michael Coughlin | School of Physics and Astronomy, UMN

- Utilised the AMPLFI package to perform **likelihood-free** inference on **neutron star** merger gravitational waves.
- Modifying the **Binary-Black Hole models from IMRPhenom** accounting for Neutron star equation of state.
- **Optimising training hyper-parameters** for robust real world performance of final model.

GRMHD Modelling of Compact Object Accretion *(Aug 2025 - Dec 2025)*

Guide : Prof. Rahul Kashyap | Physics Department, IITB | Senior Thesis Project

- Utilising AthenaK to simulate compact object accretion disks across length scales **testing scale invariance**.
- Studying the mechanism and effects of **3-D magneto-rotational instability** on angular momentum transport.
- Connecting simulations to observed data across **multiple scales** to gain insight on **disk and jet** structure.

Mapping Lunar Chemical Abundances using X-ray fluorescence *(Oct 2024 - Apr 2025)*

Placed first in the high-prep ISRO problem statement at the Inter-IIT Tech Meet | Repository

- Studied X-ray Fluorescence (XRF) and elemental detection methods related to solar flares incident on the Moon.
- Utilized Chandrayaan-2 Large Area Soft X-ray Spectrometer (**CLASS**) data to develop and implement a **novel algorithm** to detect solar flares based on strengths of elemental XRF lines and quality of their gaussian fits.
- Created the **first** high-resolution map of XRF line ratios using **GeoTIFF** files, identifying **compositional groups**.

Machine Learning for Intelligent Reflective Surfaces *(Aug 2024 - Dec 2024)*

Guide : Prof. Anshuman Kumar | Department of Physics, IITB | RnD Project

- Implemented programs to calculate the **far-field** intensity patterns of arrays of radiating elements using both **direct summation of array factors** as well as **fourier transform** based approximations.
- Implemented the **Gerchberg-Saxton(GS) iterative algorithm** to find the most suitable distributions of relative phases to produce a given beamed intensity profile.
- Trained a Pytorch based **Auto-encoder Neural Network** alternative to the GS algorithm with reduced time complexity, viable for the **real-time solution** of the inverse-design phase retrieval problem.

Honours and Scholastic Achievements

- Recieved Undergraduate Research Award-1 in recognition of work with Prof. Varun Bhalerao **2025**
- Awarded Merit-Based Scholarship by the Narottam Sekhsaria Foundation **2025**
- Awarded a change of major to Engineering Physics for being in top 10% of freshmen class GPA. **2023**
- Secured **999** rank in **JEE Advanced** exam amongst **150,000**+ candidates. **2022**
- Obtained 1036 rank and **99.89** percentile in **JEE Main** examination amongst **1 million**+ candidates **2022**
- Awarded **KVPY** Fellowship by IISc Bengaluru for securing rank of **482** amongst **50,000**+ students. **2021**

Course Projects

Observation and Isochrone Fitting of Open Clusters *(Aug 2025 - Nov 2025)*

Guide: Prof. Varun Bhalerao | Observational Astronomy Course Project

- **Proposed observations with the GROWTH-India telescope** to observe the open clusters ASCC 19 and 21.
- Performed aperture photometry on images using the aperture photometry tool (APT) as well as automatically using the **Source Exctractor** python package to build **colour-magnitude diagrams** for age determination.
- Utilised parsec isochrones and a custom **MCMC-based fitting routine**, verifying clusters are of same age.

Geometry and Observations of Thin Accretion Disks *(Jan 2025 - April 2025)*

Guide: Prof. Rahul Kashyap | Advanced Astrophysics Course Project | Presentation

- Simulated angular momentum transport using 1-D non-linear diffusion equations with different viscosity profiles of varying radial and mass dependence, **to study the formation of thin keplerian disks**.
- Utilised spectral features and accretion rates of compact object observations to **distinguish neutron stars from black holes** and their various accretion states, arriving at a unified model of accretion evolution

- Analysed timing properties of **quasi-periodic oscillations (QPOs)** from known accreting sources to compare with general relativistic predictions of ISCO, Periastron and lense-thirring frequencies.

Post-Newtonian Binary Inspiral Calculations *(Oct 2025 - Dec 2025)*

Guide: Prof. Archana Pai | Gravitational Wave Astronomy Course Project

- Performed a literature review on Post-Newtonian Inspiral Calculations upto 1PN and IMRphenom models.
- Comparatively studied features and new physics entering at every post-newtonian order.
- Created a report and presentation delivered it to an audience of peers, professors and other active researchers.

Precision Neutrino Cosmology *(Aug 2025 - Nov 2025)*

Guide: Prof. Manibrata Sen | Astro-Particle Physics Course Project

- Derived effects of neutrinos on cosmological evolution and their observable signatures which have been detected.
- Performed a comparative study of various precision detection techniques to detect the effect of neutrinos.
- Presented the evolution of the bounds on neutrino mass and energy density in recent decades.

Professional Experience

Deep Learning Methods for Depth and Elevation Estimation *(May 2026 - Aug 2026)*

SatSure Analytics | Summer Data Science Internship

- Implemented neural network based RAFT-STEREO to replace classical stereo-satellite image pipelines for **elevation estimation** using stereo satellite image pairs.
- Utilized Meta's **DINOv2 transformer model** to extract depth information from single satellite images
- **Benchmarked performance** for construction monitoring tasks against open source Google Buildings Dataset.
- Created **comparison suite** to compare deep learning approaches against **ground truth LiDAR** datasets.

Teaching and Leadership Experience

Teaching Assistant (TA) | Department of Physics, IITB *(Spring'23, Fall '24, Fall '25, Fall'26)*

Courses : Astrophysics | Classical and Quantum Mechanics | Analog Electronics | AI and Data Science

- Supervised 35 student's projects using GROWTH-India Telescope in observational astronomy course.
- Responsible for **conducting 12 tutorials**, covering problem sets on **classical and quantum physics**.
- Engaged in the logistics of correcting answer scripts and **conducting help sessions** for struggling students.
- Successfully fulfilled the role of Lab TA, guiding students through **10 analog electronics assignments**
- Guiding sophomores through 10 jupyter-based tutorials to **develop coding and statistics knowledge**.

Manager of Physics Journal Club | Department of Physics, IITB *(June 2025 - May 2026)*

- Responsible for recruiting and training a roster of physics students for biweekly student research presentations.
- Engaged in the preparation of resources for budding researchers from experiences of seniors in graduate school.

Department Academic Mentorship Programme | DAMP Mentor *(Jun 2024 - Apr 2026)*

- Responsible for **mentoring 8 juniors** and guiding them on their transition to second year in the department.
- Volunteering in the **English Learning Program** as TA, Providing one-on-one weekly sessions for **personalised help** to several mentees in order to improve their english proficiency to aid in their professional lives.
- Currently performing the duties of **Academic Rehabilitation Mentor** to help struggling juniors.

Tournament Director of the IIT Bombay Debate *(May 2024 - May 2025)*

- Entrusted with executing **Asia's biggest British Parliamentary debate** tournament by IITB's internationally successful debating society with a budget of **1 Million INR**.
- Spearheaded **negotiations and discussions** with external companies in order to obtain **sponsors** for the event.
- Organised 2 workshops held by top international debaters to facilitate skill development within the circuit.

Extracurricular Activities

- Attended ZTF summer school 2025 and 4th Time Domain and Multi-Messenger Workshop online (2025)
- Mentored 5 juniors in the Krittika Summer Project 2025 and oversaw creation of posters for national space day.
- Volunteered at IIT Bombay Techfest for the LIGO-India and Daksha stalls for outreach. (Dec 2024)
- Underwent intensive cricket training as part of the National Sports Office program. (December 2022 - June 2023)
- Won the Astromania quiz held by the astronomy club of IITB named 'Krittika' (Oct 2023)
- Active debater and community organiser with the debate club of IIT Bombay, WeSpeak (2022-present)
- Delivered a talk to first year students on GRBs as a contributor to the Physics Journal Club. (October 2023)

Relevant Coursework and Skills

Physics Theory:	Gravitational Waves, Astrofluidics, Astroparticle Physics and Cosmology, Observational Astronomy, General Relativity, Electromagnetism 1 & 2, Statistical Physics, Classical Mechanics, Quantum Mechanics I and II, Advanced Statistical Mechanics, Non-Linear Dynamics
Mathematics :	Introduction to Numerical Analysis, Quantum Information and Computing I, Differential Equations, Complex Analysis and Integral Transforms, Linear Algebra, Calculus I and II
Machine Learning :	AI and Data Science, Statistical Machine Learning and Data Mining
Software Experience :	Linux, SLURM, Kokkos, Python (PyTorch, SciPy, GWpy, pymcmc), Git, LATEX, Carta, Aperture Photometry, Mathematica, Matlab, Arduino

References

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